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CONFORMAL GRAVITY AND THE FLATNESS PROBLEM

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ABSTRACT

We present the radiation and matter-dominated cosmologies associated with fourth-order conformal Weyl gravity, a theory which is currently being explored as a candidate alternative to the standard second-order Einstein theory. The theory naturally yields an open though nonetheless recollapsing universe which can even oscillate indefinitely, which appears to possess no flatness problem, which has a cosmology which could potentially be created out of nothing, which is singularity free, and which does not appear to require any cosmological dark matter.

Subject heading: cosmology: theory

1. INTRODUCTION

Even though it is now more or less universally accepted that the masses of the elementary particles and the mass scales associated with the fundamental strong, electromagnetic, and weak interactions are generated spontaneously, it is almost as equally accepted that the Planck mass scale associated with gravity is strictly kinematical and that it should appear in the Einstein-Hilbert gravitational action simply as an intrinsic ab initio dimensionful parameter. Recognizing this to be a somewhat serious disparity, over the years a small but not insignificant set of authors (see, e.g., Adler 1982 and Zee 1983 for details and references to the previous literature) has actually entertained the possibility that gravity itself might also have some underlying dimensionless scale-invariant structure and has considered conformal Weyl gravity as such a candidate alternate gravitational theory, this being the theory which is based on local conformal invariance $[g_{\mu\nu}(x) \rightarrow \Omega^2(x) g_{\mu\nu}(x)]$ of the geometry and the unique fourth-order gravitational action

$$I_W = -\alpha \int d^4x (-g)^{1/2} C_{\lambda\mu\nu\kappa} C^{\lambda\mu\nu\kappa} , \tag{1}$$

where $C_{\lambda\mu\nu\kappa}$ is the conformal Weyl tensor and α is a purely dimensionless coefficient. In the main, the thrust of these efforts has been to try to dynamically break the conformal theory so that the Einstein theory would then emerge as its effective low-energy limit in analogy to the way the four-Fermi theory is obtained as the low-energy limit of the Weinberg-Salam theory. (While these attempts have met with varying degrees of success and difficulty, one of the problems they have encountered is that giving a vacuum expectation value to a scalar field which is conformally coupled to gravity does indeed appear to induce an effective Einstein-Hilbert action, only one with repulsive rather than attractive gravity. As we will see below, this "wrong sign" conformal coupling will actually turn out to be an advantage rather than a disadvantage cosmologically.)

As an alternative to this canonical approach to symmetry breaking, quite recently it has been suggested by Mannheim and Kazanas in a series of papers to eschew looking for an effective low-energy Einstein limit altogether, and instead to consider the fourth-order theory in and of itself in its own right, to try to find exact solutions to it, and to confront it with observation directly. To this end quite a big set of exact solutions has now been found, these being the fourth-order analogs of the exterior Schwarschild solution (Mannheim & Kazanas 1989), the interior Schwarzschild solution (Mannheim & Kazanas 1990), and the exterior Reissner-Nordstrom, the exterior Kerr, and the exterior Kerr-Newman solutions (Mannheim & Kazanas 1991a). In addition, it has also been found (Mannheim 1990) that not only does conformal invariance forbid the presence of any kinematic cosmological term at the level of the input Lagrangian (since the cosmological term is dimensionful), but also the theory possesses sufficient residual symmetry to even prevent the appearance of any induced cosmological term in a background de Sitter geometry once the conformal symmetry is broken through the nonvanishing of the vacuum expectation value of a scalar field. Conformal gravity thus naturally resolves the cosmological constant problem.

With regard to the Newtonian limit of the theory and the recovering of the classic tests of general relativity, the exact nonperturbative interior and exterior solutions associated with a static, spherically symmetric source have been obtained. The exterior metric is found (Mannheim & Kazanas 1989) to take the form

$$-g_{00} = 1/g_{rr} = 1 - \beta(2 - 3\beta\gamma)/r - 3\beta\gamma + \gamma r - kr^2 , \tag{2}$$

where β, γ, and k are three appropriate dimensionful integration constants. The matching of the interior and exterior solutions confirms the consistency of the above exterior metric while additionally relating its parameters to interior properties of the gravitating system according to the exact nonperturbative relations (Mannheim & Kazanas 1990):

$$\beta(2 - 3\beta\gamma) = \tfrac{1}{6} \int_0^R dr' f(r') r'^4 , \tag{3}$$

$$\gamma = -\tfrac{1}{2} \int_0^R dr' f(r') r'^2 , \tag{4}$$

where the function $f(r)$ is given by

$$f(r) = 3(T^0{}_0 - T^r{}_r)/4\alpha B(r) . \tag{5}$$

We see that the solution contains the required familiar exterior Schwarzschild term together with two additional potential terms. The kr^2 term represents an uninteresting background de Sitter geometry which the conformal theory can support in its vacuum even in the absence of any kinematic or induced cosmological term. The linear γr term is a new gravitational term and is a special and characteristic feature of the fourth

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order theory. For small enough γ the Weyl theory would thus appear to enjoy the static experimental successes of Einstein relativity, all of which have been obtained on solar system or smaller distance scales. Further, it was suggested (Mannheim & Kazanas 1989) that the linear term might first manifest itself phenomenologically on galactic distance scales where it could then prevent galactic rotation curves from undergoing a Keplerian falloff as a function of distance and bring them into qualitative agreement with observation without the apparent need for any dark matter. For this to occur for a galaxy, we find that the parameter γ should typically be of the order of 10^{-28} cm^{-1}, a value which interestingly is of the order of the inverse Hubble length.

As we can see, the conformal theory can thus potentially address some outstanding issues of observational astrophysics and recover the Schwarzschild solution on distance scales where the linear potential term is unimportant even while never recovering the Einstein equations of motion themselves in any limit at all, and must therefore be regarded as a currently viable theory of gravity. (For a review of its current status and the challenges that it still faces see Mannheim & Kazanas 1991b). To continue the exploration of the theory it is also necessary to confront the standard big bang cosmology, another testing ground for the Einstein theory, and this is the main objective of this paper.

2. FORMULATION OF THE THEORY

To formulate the theory we first identify the functional derivative $(-g)^{-1/2}\delta I_W/\delta g_{\mu\nu}$ as the specific gravitational rank two tensor $-2\alpha W^{\mu\nu}$ of the conformal theory (this then being the analog here of the Einstein tensor $G^{\mu\nu}$ of the standard theory). For the conformal Weyl action we find that $W_{\mu\nu}$ is given by (for details see, e.g., Kazanas & Mannheim 1991)

$$W_{\mu\nu} = W^{(2)}_{\mu\nu} - W^{(1)}_{\mu\nu}/3 , \tag{6}$$

where

$$W^{(1)}_{\mu\nu} = 2g_{\mu\nu}(R^{\alpha}{}_{\alpha})^{;\beta}{}_{;\beta} - 2(R^{\alpha}{}_{\alpha})_{;\mu;\nu} - 2R^{\alpha}{}_{\alpha}R_{\mu\nu} + g_{\mu\nu}(R^{\alpha}{}_{\alpha})^2/2 ,$$
$$W^{(2)}_{\mu\nu} = g_{\mu\nu}(R^{\alpha}{}_{\alpha})^{;\beta}{}_{;\beta}/2 + R_{\mu\nu}{}^{;\beta}{}_{;\beta} - R_{\mu}{}^{\beta}{}_{;\nu;\beta} - R_{\nu}{}^{\beta}{}_{;\mu;\beta} - 2R_{\mu\beta}R_{\nu}{}^{\beta} + g_{\mu\nu}R_{\alpha\beta}R^{\alpha\beta}/2 , \tag{7}$$

so that the gravitational equations of motion take the form

$$-4\alpha W^{\mu\nu} + T^{\mu\nu} = 0 , \tag{8}$$

where $T^{\mu\nu}$ is the conformal energy momentum tensor. Because of the conformal invariance of the theory, both $W^{\mu\nu}$ and $T^{\mu\nu}$ are kinematically traceless, a constraint which will prove central in the following.

For the matter fields we need only consider a fermion field $\psi(x)$ and a scalar field $S(x)$, the fermion field to generically represent normal matter and the scalar field to produce the symmetry breaking. For these fields the general covariant conformal matter action is given by

$$I_M = -\int d^4x(-g)^{1/2}[S^{\mu}S_{\mu}/2 + \lambda S^4 - S^2R^{\mu}{}_{\mu}/12 + i\bar{\psi}\gamma^{\mu}(x)(\partial_{\mu} + \Gamma_{\mu}(x))\psi - hS\bar{\psi}\psi] , \tag{9}$$

where $\Gamma_{\mu}(x)$ is the fermion spin connection and h and λ are dimensionless coupling constants. With this action the matter field equations of motion take the form

$$i\gamma^{\mu}(x)[\partial_{\mu} + \Gamma_{\mu}(x)]\psi - hS\psi = 0 , \tag{10}$$

and

$$S^{\mu}{}_{;\mu} + SR^{\mu}{}_{\mu}/6 - 4\lambda S^3 + h\bar{\psi}\psi = 0 , \tag{11}$$

while the conformal energy-momentum tensor is given by

$$T_{\mu\nu} = i\bar{\psi}\gamma_{\mu}(x)[\partial_{\nu} + \Gamma_{\nu}(x)]\psi + 2S_{\mu}S_{\nu}/3 - g_{\mu\nu}S^{\alpha}S_{\alpha}/6 - SS_{\mu;\nu}/3 + g_{\mu\nu}SS^{\alpha}{}_{;\alpha}/3 - S^2(R_{\mu\nu} - g_{\mu\nu}R^{\alpha}{}_{\alpha}/2)/6 - g_{\mu\nu}\lambda S^4 . \tag{12}$$

Thus when the scalar field acquires a nonvanishing vacuum expectation value, the fermion acquires a mass $m = hS$.

Because of the conformal invariance of the theory, should the solutions to the theory yield a spacetime-dependent value for $S(x)$, we can always remove this spacetime-dependence by a conformal transformation and bring $S(x)$ to a spacetime-independent constant (i.e., the physics is in this constant being nonzero), so with no loss of generality we may replace equations (11) and (12) by

$$SR^{\mu}{}_{\mu}/6 - 4\lambda S^3 + h\bar{\psi}\psi = 0 , \tag{13}$$

and

$$T_{\mu\nu} = i\bar{\psi}\gamma_{\mu}(x)[\partial_{\nu} + \Gamma_{\nu}(x)]\psi - S^2(R_{\mu\nu} - g_{\mu\nu}R^{\alpha}{}_{\alpha}/2)/6 - g_{\mu\nu}\lambda S^4 . \tag{14}$$

Because of equation (13), $T_{\mu\nu}$ may also be written in the form

$$T_{\mu\nu} = i\bar{\psi}\gamma_{\mu}(x)[\partial_{\nu} + \Gamma_{\nu}(x)]\psi - g_{\mu\nu}hS\bar{\psi}\psi/4 - S^2(R_{\mu\nu} - g_{\mu\nu}R^{\alpha}{}_{\alpha}/4)/6 \tag{15}$$

to make its tracelessness manifest.

With this last form for the energy-momentum tensor it is also possible to establish the perfect fluid form needed for cosmology. Specifically, if we momentarily restrict the theory to flat spacetime, the quantization of the theory is straightforward and yields one particle plane wave eigenstates $|k\rangle$ of four momentum $k^{\mu} = \{E_k = (\boldsymbol{k}^2 + m^2)^{1/2}, \boldsymbol{k}\}$. For these states we obtain (Mannheim 1990) the matrix elements

$$\langle k|\int d^3xT_{00}|k\rangle = E_k - m^2/4E_k \tag{16}$$

$$\langle k|\int d^3xT^{i}{}_{i}|k\rangle = \boldsymbol{k}^2/E_k + 3m^2/4E_k = -\langle k|\int d^3xT^{0}{}_{0}|k\rangle \tag{17}$$

so that the tracelessness property is still manifest. In equations (16) and (17) we note that the energy is reduced from the usual one particle E_k term by the vacuum contribution $-m^2/4E_k$ while the usual kinematic $\boldsymbol{k}^2/E_k$ pressure is augmented by a corresponding vacuum contribution. These vacuum contributions (which are akin to the Poincare stresses of a completely electrodynamical electron) thus appear when the mass is dynamical and serve to maintain the tracelessness of $T_{\mu\nu}$, to thus show that the tracelessness of $T_{\mu\nu}$ does not in fact necessitate masslessness even while it does entail the vanishing of kinematic masses. An incoherent averaging over these plane wave states then yields a perfect fluid, two in fact, one being a standard kinematic fluid

$$T^{\rm kin}_{\mu\nu} = (\rho + p)U_{\mu}U_{\nu} + pg_{\mu\nu} \tag{18}$$

coming from the averaging over the fermion kinetic energy term, with the other (which we denote by $T^{\rm sc}_{\mu\nu}$) being due to the self consistent $-g_{\mu\nu}hS\bar{\psi}\psi/4$ term, and since $T^{\rm sc}_{\mu\nu}$ serves to main-

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tain tracelessness in the fermion sector it must take the form

$$T^{sc}_{\mu\nu} = -g_{\mu\nu}(3p - \rho)/4 , \quad (19)$$

where ρ and p are the kinematic energy density and pressure of $T^{kin}_{\mu\nu}$. Even though an interplay between the two fluids is necessary for the tracelessness of the energy-momentum tensor, we note that no such interplay is needed in order to enforce its covariant conservation. Specifically, the fermion kinetic energy term contribution to $T_{\mu\nu}$ is covariantly conserved all by itself when $S(x)$ is constant simply because of the fermion equation of motion. (This can also be seen directly from eq. [14] since $S(x)$ is a constant and the Einstein tensor anyway obeys the Bianchi identities.) Thus the kinematic ρ and p of equation (18) obey the standard covariant conservation equations of a Friedmann cosmology and the standard equations of state even in the presence of the spontaneous mass-generation mechanism. Of course in the high-temperature regime $T^{sc}_{\mu\nu}$ is nonleading and the fermion kinetic energy term then acts like a regular $\rho = 3p$ blackbody fluid in this limit, so that the self-consistent term is anway only relevant in the matter-dominated case. Having now set up the model we proceed next to see what sort of cosmology it could yield.

3. CONFORMAL COSMOLOGY

In considering cosmology we note immediately that a Robertson-Walker metric

$$ds^2 = dt^2 - R^2(t)[dr^2/(1 - kr^2) + r^2\, d\Omega] \quad (20)$$

is conformal to a flat metric (and would remain so under any conformal transformation that would be needed to bring the scalar field to its constant value). Consequently, in such a geometry both the rank four Weyl tensor $C_{\lambda\mu\nu\kappa}$ of equation (1) and the rank two gravitational tensor $W_{\mu\nu}$ of equation (6) vanish identically. In conformal gravity then the only possible Robertson-Walker cosmologies are those in which the energy-momentum tensor is everywhere zero. (Any cosmological inhomogeneities such as those due to large-scale structure would then represent a perturbation around $T_{\mu\nu} = 0$.) As we shall see, the local vanishing of $T_{\mu\nu}$ in a Robertson-Walker geometry is not an empty statement, since in the presence of normal matter it can be achieved nontrivially by an interplay between the various fields of the theory. Because of the maximal three space symmetry of the geometry and the tracelessness of $T_{\mu\nu}$, the only nontrivial component of equation (8) is the time-time one, and in the metric of equation (20) equation (14) yields

$$\rho + \lambda S^4 + S^2(\dot{R}^2 + k)/2R^2 = 0 , \quad (21)$$

while equations (13) and (19) yield

$$3p - \rho - 4\lambda S^4 - S^2(R\ddot{R} + \dot{R}^2 + k)/R^2 = 0 , \quad (22)$$

and the covariant conservation of $T^{kin}_{\mu\nu}$ yields

$$3p + 3\rho + R\, d\rho/dR = 0 . \quad (23)$$

Explicit calculation then shows that equation (22) is not independent of equations (21) and (23), and with ρ being taken as a function of R, equation (21) may be integrated by quadrature to yield

$$\int dt = \int dR[-k - 2R^2(\rho S^{-2} + \lambda S^2)]^{-1/2} . \quad (24)$$

Solutions to equation (24) are readily found in the radiation-dominated era where $\rho = 3p = A/R^4$ (A positive) and the matter-dominated one where $p = 0$, $\rho = A/R^3$ (A positive), and their structure depends only on the relative signs of the terms in its integrand as needed to keep the argument of the square root positive. Indeed, we note immediately that if λ is indeed positive as is generally thought to be the case because of stability arguments, it then follows that k must be strictly negative (even if λ just happens to be zero in fact), so that the model only supports an open universe once A is nonzero, i.e., once ordinary matter is present. (Even if for some unknown reason λ just happened to be negative, it would have to be extremely large before it could overwhelm the contribution of the energy density ρ and allow k to take other values.)

For the radiation-dominated era the integration of equation (24) is straightforward, with $\dot{R}^2$ vanishing whenever there are real solutions to the equation

$$R^2 = -k/c \pm [(k^2S^2 - 4Ac)/S^2c^2]^{1/2} \quad (25)$$

(c denotes the quantity $4\lambda S^2$). When c is positive the integration of equation (24) yields

$$R^2 = \frac{-k}{c} - \left(\frac{k^2S^2 - 4Ac}{c^2S^2}\right)^{1/2} \cos\,[(2c)^{1/2}t] . \quad (26)$$

In this solution the radius $R(t)$ can never vanish since the parameter c is nonzero, so this universe has a finite minimum size which for simplicity is chosen to occur at $t = 0$. Hence the cosmology is singularity-free, though of course it is still very hot near to this minimum radius since the quantity $TR(t)$ is constant for a blackbody. As we can also see, this open universe expands and eventually recollapses and thus stands in marked contrast to the open universe of the standard Einstein theory which never recollapses at all. Since the minimum radius is presumably macroscopic, microscopic quantum physics should not be able to influence the universe much at its minimum size, and hence the universe should be able to go right through its minimum and on into the next cycle. Thus we obtain a singularity-free, permanently oscillating eternal universe with each cycle taking a time $\pi(2/c)^{1/2}$. (With regard to the horizon structure of the universe, unfortunately the horizon associated with eq. [26] is an elliptic integral and for the moment we are unable to compare it with the size of the universe when it is at minimum radius.)

Should the parameter λ just happen to vanish, the solution to equation (24) would take the form

$$R^2 = -kt^2 - 2A/kS^2 \quad (27)$$

after setting $R^2 = -2A/kS^2$ at $t = 0$, so this open universe does not recollapse. Finally, for completeness we note that if c is negative the solution takes the form

$$R^2 = \frac{-k}{c} + \left(\frac{k^2S^2 - 4Ac}{c^2S^2}\right)^{1/2} \cosh\,[(-2c)^{1/2}(t - t_0)] , \quad (28)$$

where t_0 is an integration constant. This universe could then possibly have k positive for some choices of the parameters and still not collapse. Thus in our theory the time-dependence of $R(t)$ is not correlated with the sign of the spatial curvature, but rather with the sign of the quartic coupling contant λ, a sign which is already thought to be known.

For the matter-dominated case the integration of equation (24) does not appear to be expressible in terms of elementary functions in general. However, we have found explicit solutions in two specific cases. For $c = 0$ (which requires k negative) we obtain the relation

$$[R(R - a)]^{1/2} + a \ln\,[2R^{1/2} + 2(R - a)^{1/2}] = (-k)^{1/2}(t - t_0) , \quad (29)$$

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(where $a = -2A/kS^2$); while for $k = 0$ (which requires c to take unphysical negative values) we obtain

$$R^{3/2} = (-A/\lambda S^4)^{1/2} \cosh [3S(-\lambda/2)^{1/2}(t - t_0)] . \quad (30)$$

Having established the general structure of our solutions, it now becomes extremely important to make a detailed study of the phenomenological implications of the model, especially to find out if there is some range of parameters of the theory for which the constraints due to early universe nucleosynthesis can be met. This is of course an urgent challenge to the theory which however is beyond the scope of this short note.

In order to appreciate the difference between our approach and the standard second-order one it is useful to compare our equation (21), viz.,

$$\dot{R}^2 + k = -\frac{2R^2\rho}{S^2} - 2R^2\lambda S^2 \quad (31)$$

with the standard Friedmann universe equation

$$\dot{R}^2 + k = 8\pi G\rho R^2/3 . \quad (32)$$

As we can see the central difference is that the right-hand side of equation (31) is negative while that of equation (32) is positive. Because of the positivity of its right-hand side, equation (32) requires an incredible fine tuning, the flatness problem, which is usually resolved in the standard model by appealing to the inflationary universe scenario (Guth 1981). However, this scenario does not appear to be needed for our equation (31), since this time two large quantities would only need to add rather than subtract in order to equal k. Our equation (31) would only appear in the standard model if the Einstein-Hilbert action were to have a repulsive gravitational constant, which is of course forbidden in the Einstein theory since the sign of G is already fixed to be positive from the theory's Newtonian limit. In our theory the sign of the Newtonian limit is fixed by equations (3) and (5) which are only nontrivial in the explicit presence of inhomogeneities; and since $T^0{}_0 - T^r{}_r = -(\rho + p)$ for ordinary matter, the sign of the gravitational field put out by the Sun depends on the sign of the fundamental dimensionless coupling constant α given in the fundamental action of equation (1). (Of course a priori we do not know what this sign is, but then of course the sign of G is not explained in the second-order theory either. However, since α is dimensionless it may in principle actually prove to be calculable by some nonperturbative eigenvalue of the type envisaged for the fine-structure constant.) Since our cosmological solution of equation (26) is a pure vacuum solution (so that the associated universe could even be created out of nothing), the analysis leading to it is independent of the sign of α in the gravitational equations of motion of equation (8), so the fact that the coefficient of the Einstein tensor is negative in the energy-momentum tensor of equation (14) (thereby permitting the nontrivial cancellations needed to enforce $T_{00} = 0$ to actually occur) does not mean that the theory leads to a repulsive Newtonian limit. The two issues are simply decoupled in the conformal theory.

With this elimination of the flatness problem which is now seen to be purely a property of the structure of the equations of motion of the second-order theory rather than something generic to cosmology, it would thus appear to us that a search for cosmological dark matter may no longer be warranted (except perhaps to show that it is not there), and indeed since, as noted above, the conformal theory also has the capability of eliminating the need for galactic dark matter as well, it may actually turn out to be the case that the observed luminous matter is the major constituent of the universe after all.

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